

Reviewer Pack — Minimal Hodge Decomposition Complexity of Tropical Schemes v...

Entry a352496a80e9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Hodge Decomposition Complexity of Tropical Schemes vs. DPLL Proof Length

Entry ID: a352496a80e9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-04 00:45:29 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Hodge Theory) **Field B** (complexity object): DPLL Search Trees

Statement:

For every n -variable CNF φ , the minimal Hodge decomposition complexity ($\text{hdc}(\varphi)$) of its associated tropical scheme is linearly correlated with its DPLL proof length, such that $\text{hdc}(\varphi) = \Theta(L(\varphi))$, where $L(\varphi)$ is a function of n .

Rationale (proposer's reasoning):

Tropical geometry provides a combinatorial representation of complex algebraic varieties, which can be used to study the complexity of computing solutions to polynomial equations. By leveraging the Hodge decomposition in tropical geometry, we might uncover new structures that influence DPLL proof length, potentially revealing deeper connections between geometric properties and algorithmic complexity.

Taxonomy category: TROPICAL_GEOMETRY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 78a391d128432402

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the correlation coefficient of $\text{hdc}(\varphi)$ and $L(\varphi)$ from at least 30 independent n -variable CNF φ instances is ≥ 0.8 , with $\text{hdc}(\varphi) = \Theta(L(\varphi))$ for all φ .

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 3 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "minimal Hodge decomposition complexity" AND "tropical geometry" AND "DPLL proof length" - "Hodge theory" IN TITLE AND DPLL IN TITLE - "tropical scheme" AND CNF AND "proof length"

Top relevant hits considered: - [http://arxiv.org/abs/1810.12657v1] The effect of a country's name in the title of a publication on its visibility and citability - [http://arxiv.org/abs/cs/0207003v1] Analysis of Titles and Readers For Title Generation Centered on the Readers - [http://arxiv.org/abs/1004.2719v1] Is This a Good Title?

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=17.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2**n):
            clause = [random.choice([1, -1]) * (i + 1) for i in range(n)]
            if sum(clause) > 0:
                clauses.append(clause)
        return clauses

    def dpll(cnf):
        def solve(literals):
            if not cnf:
                return literals
            unit_clauses = [c[0] for c in cnf if len(c) == 1]
            if unit_clauses:
                literal = unit_clauses[0]
                new_cnf = [c for c in cnf if literal not in c and -literal not in c]
                return solve(literals + [literal])

            p = random.choice([l for l in range(1, n+1) if l not in literals and -l not in literals])
            new_cnf_true = [c for c in cnf if p not in c and -p not in c]
            new_cnf_false = [c for c in cnf if -p not in c and p not in c]

            result_true = solve(literals + [p])
```

```

        if result_true is not None:
            return result_true
        else:
            return solve(literals + [-p])

n = len(cnf[0]) // 2
return solve([])

def hdc(cnf):
    # Placeholder for Hodge decomposition complexity calculation
    # This is a dummy implementation and should be replaced with actual computation
    return len(cnf)

n = random.choice([5, 10, 15, 20, 30, 40])
cnf = generate_cnf(n)
hdc_value = hdc(cnf)
proof_length = dpll(cnf)

if proof_length is None:
    return {
        "metric_name": "hdc_vs_dpll",
        "metric_value": 0,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": False,
        "counterexample": "DPLL solver did not terminate"
    }

return {
    "metric_name": "hdc_vs_dpll",
    "metric_value": hdc_value / proof_length,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": False,
    "counterexample": ""
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2*i + 1 for i in range(5, 30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std=0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std=0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])

```

```
print(f"RESULT: FALSIFIED counterexample=\"\" first_failing_seed={first_failing_seed}")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f3e5af4.py", line 87, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f3e5af4.py", line 60, in run_trial
    proof_length = dpll(cnf)
                  ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f3e5af4.py", line 50, in dpll
    return solve([])
           ^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f3e5af4.py", line 43, in solve
    result_true = solve(literals + [p])
                  ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f3e5af4.py", line 43, in solve
    result_true = solve(literals + [p])
                  ^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f3e5af4.py", line 43, in solve
    result_true = solve(literals + [p])
                  ^^^^^^^^^^^^^^^^^^^^^
  [Previous line repeated 7 more times]
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_7f3e5af4.py", line 39, in solve
    p = random.choice([l for l in range(1, n+1) if l not in literals and -l not in literals])
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/random.py", line 347, in choice
    raise IndexError('Cannot choose from an empty sequence')
IndexError: Cannot choose from an empty sequence
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed before producing any data, which means that the correlation coefficient could not be calculated to verify the conjecture. | next: Re-run the test with proper error handling and ensure it completes without crashing to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	17226
2	preregistration	ollama_remote	glm4:latest	0	0	9425
3	novelty	ollama_remote	glm4:latest	0	0	9774
4	novelty	ollama_remote	glm4:latest	0	0	9314
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14611
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	88852
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15896
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	39443
9	judge	ollama_remote	glm4:latest	0	0	40442

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 244983 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/a352496a80e9.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/a352496a80e9.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/a352496a80e9.tar.gz` (if generated)